

**WHERE
INNOVATION
IS KEY**

**TIBCO BusinessWorks
SmartRouting**
TIBCO Hackaton 2026

Challenge

Scenario:

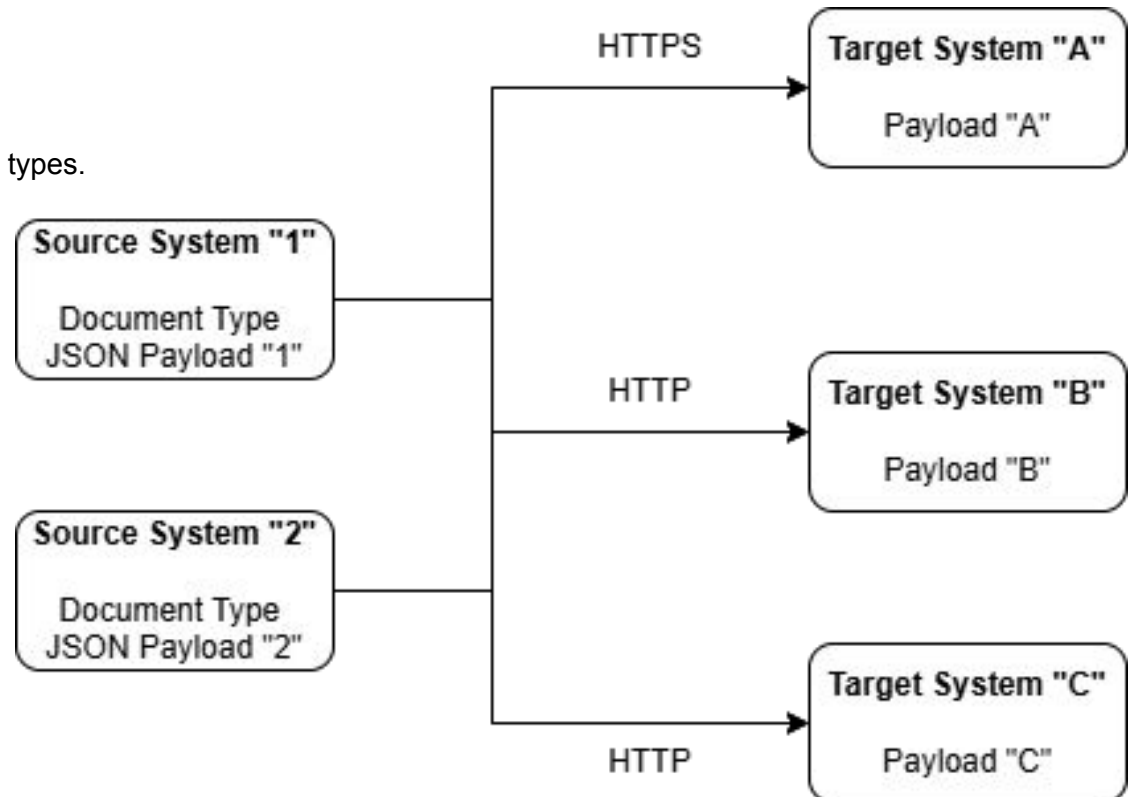
- Different source systems need to share the data about certain documents.
 - a. System "1" and System "2" both send the same document type (e.g. Purchase Order) but each one with its own payload
- A document type from a source system can have multiple target systems and each target expects a different JSON format.
 - a. System "A", System "B" and System "C" must receive the document sent by the source systems, each one with its own payload and its own protocol.
- Integrations are numerous and often point-to-point.

Challenges:

- High complexity of maintaining point-to-point integrations
- Source and target systems tightly coupled
- Effort (time/economics) required to add new sources/new targets/new document types.

Goal:

- Create a generic, secure, and reusable routing mechanism
- Standardize payload transformations
- Reduce integration effort and accelerate time-to-market



Solution Overview

SmartRouting Solution:

- REST API accepts any JSON payload + path parameters: **source system** & **document type**
- Validates if the source system is authorized to send the document type
- Retrieves target systems from configuration table
- Performs payload transformations using JOLT specification
- Dispatches transformed JSON via HTTP/HTTPS

Transformation Sequence:

1. Source JSON → Common Data Model JSON (based on document type)
2. Common Data Model JSON → Target System JSON

Result: Source and target systems remain decoupled, enabling scalable integrations.

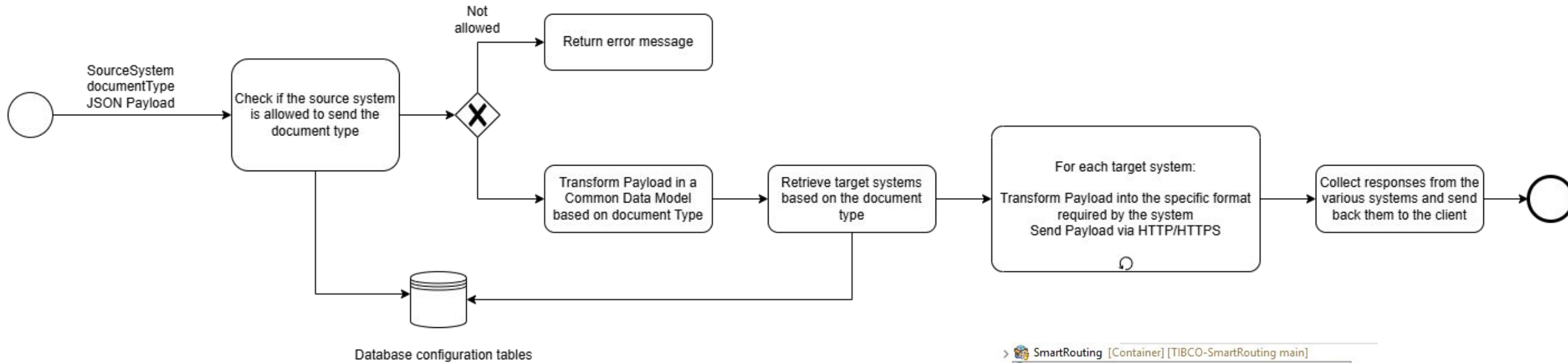
Use Cases:

- Multi-system document routing
- Cross-department or partner integrations
- Data normalization between heterogeneous systems
- Rapid onboarding of new systems and document types

Key Benefits:

- Generic & reusable API for multiple source/target systems
- Accelerates integrations and reduces development effort
- Decouples source and target via Common Data Model
- Easy configuration-based onboarding for new systems
- Secure: ensures only authorized sources send documents
- Flexible transformation management with JOLT specs

Technical implementation - Diagram flow

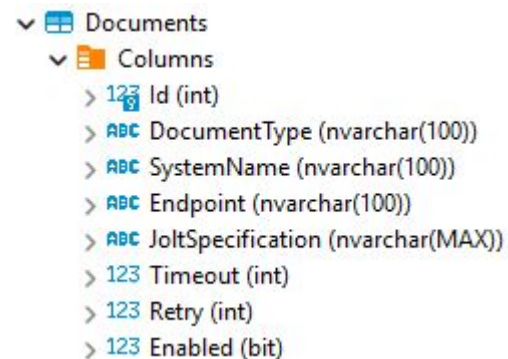
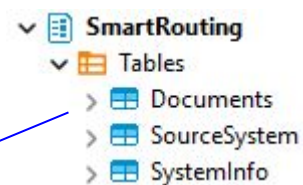


- > SmartRouting [Container] [TIBCO-SmartRouting main]
- ✓ SmartRouting.module [Container] [TIBCO-SmartRouting main]
 - ✓ Processes
 - ✓ integration
 - > JMSMessageDecoupling.bwp
 - > retrieveTarget.bwp
 - > SendHTTP.bwp
 - > SendHTTPS.bwp
 - ✓ main
 - > AsyncDispatcher.bwp
 - > Ingestion.bwp
 - ✓ operation
 - > Orchestrator.bwp
 - > SourceValidationAndTransform.bwp
 - > TargetTransformationAndSend.bwp
 - > Service Descriptors
 - > Resources
 - > Schemas
 - > Policies
 - > Module Descriptors
 - > build.properties

JOLT Specification

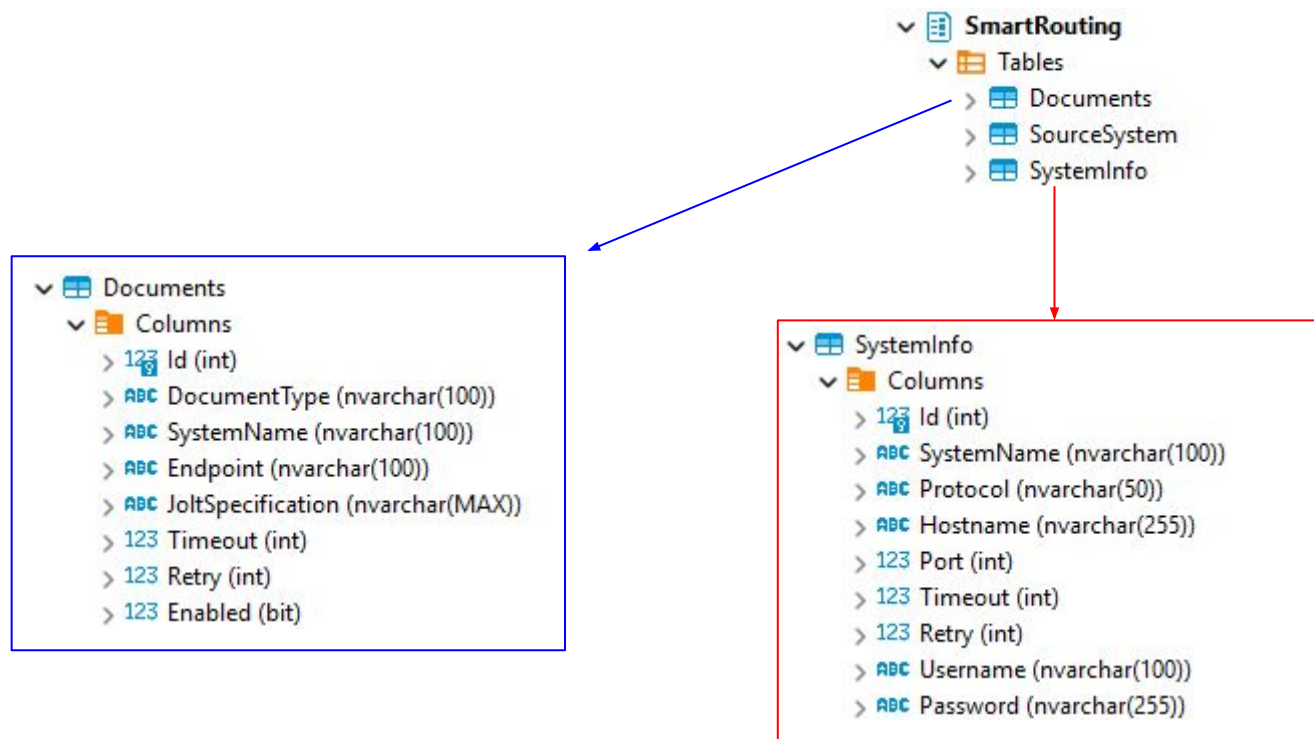
Payload Source "1"	JOLT Transformation	Common Data Model	JOLT Transformation	Payload Target System "A"
<pre>{ "orderid": 123, "line": [{ "itemCode": "001", "QUANTITY": 3, "customerName": "John", "customerStreet": "madison square" }, { "itemCode": "002", "QUANTITY": 6, "customerName": "Max", "customerStreet": "Rome Square" }] }</pre>	<pre>[{ "operation": "shift", "spec": { "orderid": "orderNumber", "line": { "**": { "itemCode": "item[&1].code", "QUANTITY": "item[&1].quantity", "customerName": "item[&1].shipTo.name", "customerStreet": "item[&1].shipTo.street" } } } }]</pre>	<pre>{ "orderNumber": 123, "item": [{ "code": "001", "quantity": 3, "shipTo": { "name": "John", "street": "madison square" } }, { "code": "002", "quantity": 6, "shipTo": { "name": "Max", "street": "Rome Square" } }] }</pre>	<pre>[{ "operation": "shift", "spec": { "orderNumber": "title", "item": { "0": { "code": "body", "shipTo": { "name": "userId" } } } } }]</pre>	<pre>{ "title": 123, "body": "001", "userId": "John" }</pre>

Database tables structure



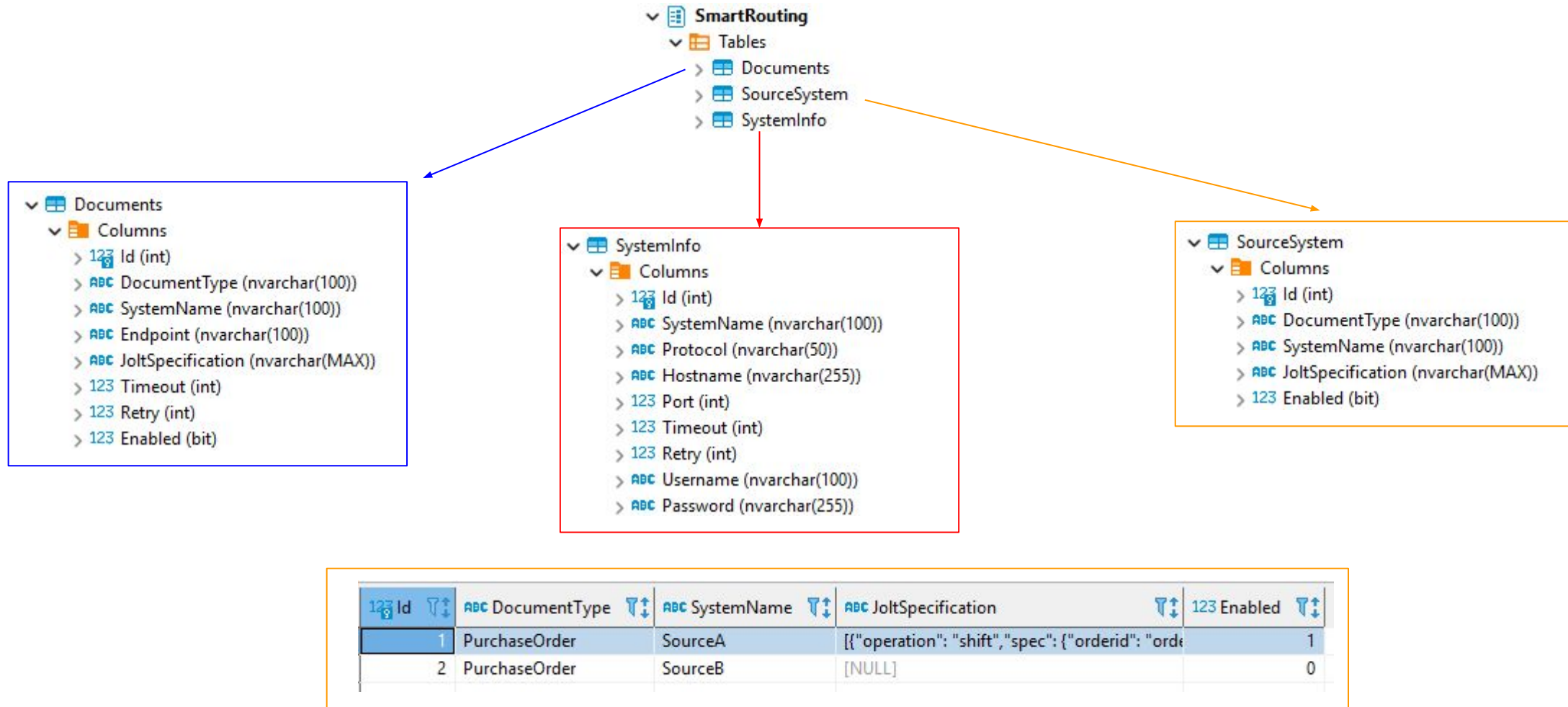
123 Id	ABC DocumentType	ABC SystemName	ABC Endpoint	ABC JoltSpecification	123 Timeout	123 Retry	123 Enabled
1	PurchaseOrder	SAP	/posts	[{"operation": "shift", "spec": {"orderNumber": "titl	5	5	1
2	PurchaseOrder	DummySystem	/erp/po	[{"operation": "shift", "spec": {"orderNumber": "titl	1	1	1
3	Delivery	DummySystem	/delivery/v1	[{"operation": "shift", "spec": {"orderNumber": "titl	3	3	0

Database tables structure



Id	SystemName	Protocol	Hostname	Port	Timeout	Retry	Username	Password
1	SAP	HTTPS	jsonplaceholder.typicode.com	443	30	5	[NULL]	[NULL]
2	SystemA	HTTP	dummyhostname.com	1,443	10	2	userA	pass
3	DummySystem	HTTP	dummyhostname.com	1,443	10	2	userA	pass

Database tables structure



Live Test

Optional Configuration Parameters

To enhance integration flexibility, the solution supports two additional query parameters, both optional and not-exclusive

- **targetSystem (optional):**
 - Sends the data only to a specific target system
 - Useful for resubmissions or targeted delivery
- **fireAndForget (optional):**
 - Source system does not wait for responses from target systems
 - Flow immediately returns an acknowledgement
 - Dispatch decoupled via **JMS queue**, enabling asynchronous processing (Messaging system required)

An example of a complete endpoints using both the parameters would be:

```
/smartrouting/{sourceSystem}/{documentType}/v1?fireAndForget=1&targetSystem={ParticularTargetSystem}
```

Benefit:

- Increased control over routing and system load
- Supports resubmissions and asynchronous integration scenarios

Thank you for the attention

KEYPARTNER
DIGITAL



Federico Preiato
Solution Architect



www.keypartner.com



info@keypartner.com



[keypartner-italia](https://www.linkedin.com/company/keypartner-italia)



[Keypartneritalia](https://www.instagram.com/Keypartneritalia)



Roma

Viale Avignone, 12
00144 Roma (RM)



Milano

Via Caldera, 21
20153 Milano (MI)



Termoli

Via Corsica, 153
86039 Termoli (CB)